



Management Performance:

Hea 14 Thermal comfort



Credits

No Minimum Standard

Aim

To ensure that asset users are thermally comfortable in occupied spaces.

Question

Is the thermal comfort of asset users in occupied spaces regularly monitored?

Credits	Answer	Select all answer options that apply (only one of C or D can be selected)
0	A.	Question not answered
0	B.	No
2	C.	Yes, through temperature measurement
4	D.	Yes, through detailed thermal comfort measurement and analysis
4	E.	Yes, through occupant thermal satisfaction surveys

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	The temperature within a representative number of the asset's occupied spaces is measured and monitored during operational hours throughout the year. Measurement can be performed using real-time temperature sensor networks or systems (e.g. building management system readings), temperature data loggers or spot measurements.	C
2.	The content and format of occupant thermal satisfaction surveys is in accordance with the Methodology section. Surveys should be completed at least annually and all asset users, or a representative sample of asset users, must be offered the opportunity to complete the survey.	E
3.	A reasonable response rate should be achieved for the survey. A response rate of at least 35% is considered reasonable. Where there are fewer than 45 occupants. Assessors should use their judgement to determine a reasonable response rate.	E

Criterion	Assessment criteria	Applicable Answer
4.	Thermal comfort of asset users in occupied space is evaluated using a detailed thermal comfort measurement and analysis method. Compliant thermal comfort methods include: - ANSI/ASHRAE Standard 55-2017 – Thermal Environmental Conditions for Human Occupancy - As a minimum, achieve Informative Appendix L PMV Model criteria for passing or Adaptive Model criteria for passing. - ISO 7730:2005 Ergonomics of the thermal environment - Analytical determination and interpretation of thermal comfort using calculation of the PMV and PPD indices and local thermal comfort criteria - As a minimum, achieve Annex A Category C criteria for PPD, PMV and local discomfort. - EN 16798-1:2019 Energy performance of buildings - Ventilation for buildings - Part 1: Indoor environmental input parameters for design and assessment of energy performance of buildings addressing indoor air quality, thermal environment, lighting and acoustics - Module M1-6 - As a minimum, achieve Annex B Category III criteria for mechanically heated and cooled buildings or for buildings without mechanical cooling systems. Alternatively, where local standards have similar requirements to any of the above standards, the local standard may be used to demonstrate compliance with this criterion.	D
5.	Procedures must be in place that cover how issues and feedback identified through the temperature and thermal comfort measurement and/or occupant thermal satisfaction surveys are addressed. The results and any potential improvement measures should be presented to and considered by senior management.	All

Methodology

Occupant thermal satisfaction surveys

ANSI/ASHRAE Standard 55-2017 - Thermal Environmental Conditions for Human Occupancy provides an example thermal satisfaction survey. The following is an adapted version of the survey which asset owners or managers can use as part of demonstrating compliance with this issue.

Survey question	Answers
1. How satisfied are you with the temperature in your space?	- Very satisfied - Satisfied - Somewhat satisfied - Neither satisfied nor dissatisfied

Survey question	Answers
	• Somewhat dissatisfied • Dissatisfied • Very dissatisfied
2. If you are dissatisfied with the temperature in your space, which of the following contribute to your dissatisfaction: a. In warm/hot weather, the temperature in my space is: b. In cool/cold weather, the temperature in my space is:	• Always too hot • Often too hot • Occasionally too hot • Occasionally too cold • Often too cold • Always too cold
3. If you are dissatisfied with the temperature in your space, which of the following contribute to your dissatisfaction: a. When is this most often a problem (check all that apply)?	• Morning (before 1100) • Midday (1100-1400) • Afternoon (1400-1700) • Evening (after 1700) • Weekends or holidays • Monday mornings • No specific time • Always • Other
4. If you are dissatisfied with the temperature in your space, which of the following contribute to your dissatisfaction: a. How would you describe the source of this discomfort (check all that apply)?	• Humidity too high (damp) • Humidity too low (dry) • Air movement too high • Air movement too low • Incoming sun • Heat from office equipment • Draughts from windows • Draughts from vents • My area is hotter or colder than other areas • Thermostat inaccessible • Thermostat is adjusted by other people • Clothing policy is not flexible • Heating/cooling system does not respond quickly enough to the thermostat • Hot or cold surrounding surfaces (floor, ceiling, walls or windows) • Deficient window (not operable) • Other

Survey question	Answers
5. If you are dissatisfied with the temperature in your space, which of the following contribute to your dissatisfaction:	Please describe any other issues related to being too hot or too cold in your space.

Information collected as part of wider occupant satisfaction surveys, including those performed in accordance with the Man 02 Management engagement and feedback assessment issue requirements, may also be used to demonstrate compliance with the criteria. For example, there are numerous commercially available occupant satisfaction evaluation tools that include thermal comfort related content (e.g. BUS methodology) whose outputs will be relevant to this assessment issue.

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Procedure and records for addressing thermal comfort related feedback and issues.
1, 2, 5	Occupant satisfaction surveys, response rate, analyses and associated records.
4	Physical measurement data, analyses and records.
5	Documentation, e.g. meeting minutes, showing senior management has reviewed the results of the survey and any potential improvement measures.

Definitions

Occupied space:

A room or space within the asset that is likely to be occupied for 30 minutes or more by an asset user. For the purpose of this issue, the definition excludes the following:

- a) Atria or concourses
- b) Entrance halls or reception areas
- c) Ancillary space, e.g. circulation areas, storerooms and plant rooms
- d) Operational or storage areas in industrial assets.